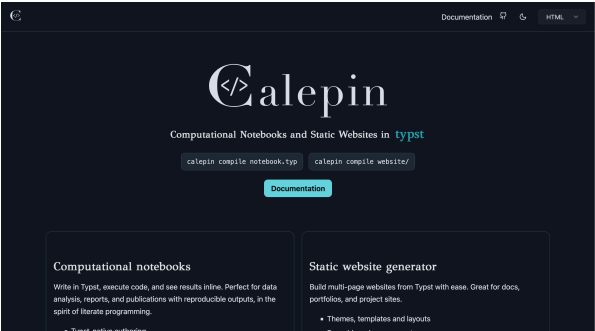
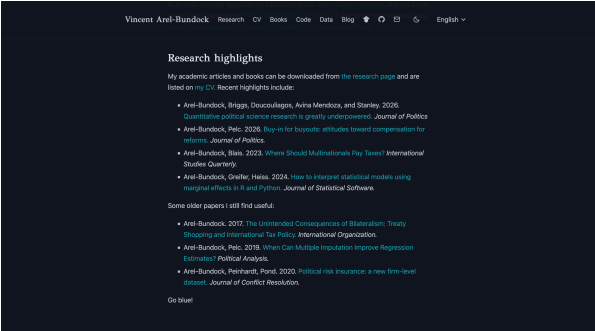


Themes

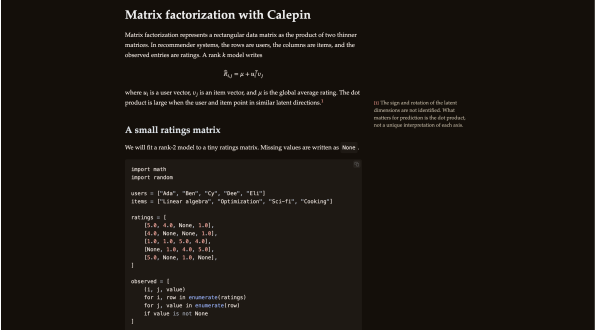
Themes control how *Calepin* renders two types of content: websites and notebooks (HTML or PDF). A website theme can provide MiniJinja HTML templates, partials, CSS, JavaScript, and notebook themes add typst-side template support for HTML and PDF outputs. The default theme is called calepin.



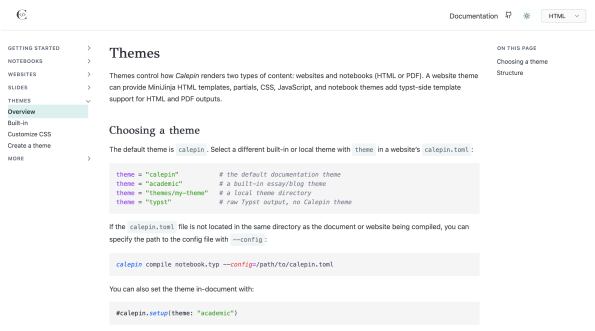
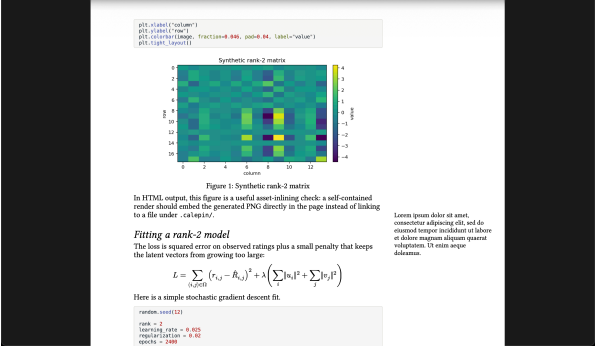
Calepin website theme in dark mode



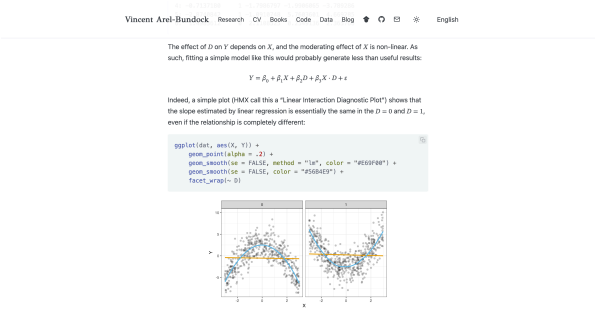
Academic website theme in dark mode



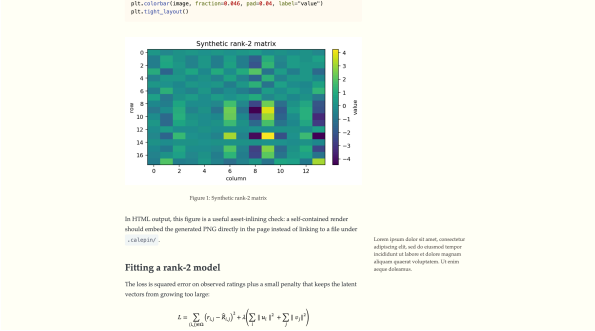
Tuftes notebook theme in dark mode



Calepin website theme in light mode



Academic website theme in light mode



Tuftes notebook theme in light mode

MATRIX FACTORIZATION WITH CALEPIN

Matrix factorization represents a rectangular data matrix as the product of two thinner matrices. In recommender systems, the rows are users, the columns are items, and the observed entries are ratings. A rank k model writes

$$\hat{R}_{i,j} = \mu + u_i^T v_j$$

where u_i is a user vector, v_j is an item vector, and μ is the global average rating. The dot product is large when the user and item point in similar latent directions.¹

A small ratings matrix

We will fit a rank-2 model to a tiny ratings matrix. Missing values are written as `None`.

```
import math
import random

users = ["Ada", "Ben", "Cy", "Dee", "Eli"]
items = ["Linear algebra", "Optimization", "Sci-Fi", "Cooking"]

ratings = [
    [5.0, 4.0, None, 1.0],
    [4.0, None, None, 1.0],
    [1.0, 1.0, 5.0, 4.0],
    [None, 1.0, 4.0, 5.0],
    [5.0, None, 1.0, None],
]

observed = [
    [i, j, value]
```

¹The sign and rotation of the latent dimensions are not identified. What matters for prediction is the dot product, not a unique interpretation of each axis.

Choosing a theme

Select a different built-in or local theme with `theme` in a website's `calepin.toml`:

```
theme = "calepin"      # the default documentation theme
theme = "academic"     # a built-in essay/blog theme
theme = "themes/my-theme" # a local theme directory
theme = "typst"        # raw Typst output, no Calepin theme
```

If the `calepin.toml` file is not located in the same directory as the document or website being compiled, you can specify the path to the config file with `--config`:

```
calepin compile notebook.typ --config=/path/to/calepin.toml
```

You can also set the theme in-document with:

```
#calepin.setup(theme: "academic")
```

When several theme settings are present during a compile, the document setting wins, then `calepin.toml`, then the default theme.

Structure

A theme can provide templates for website pages, single-document HTML, and Typst-level notebook rendering:

```
themes/my-theme/
theme.toml      # theme metadata
layouts/
  webpage.html  # layout for website pages
  notebook.html # layout for a single notebook rendered to HTML
  landing.html  # optional page-specific website layout
partials/       # MiniJinja fragments
styles/         # CSS files
scripts/        # JavaScript files
notebook.typ.jinja # Typst notebook template for PDF (and SVG)
```

The files in `layouts/` and `partials/` use the MiniJinja template language.